

The Metaphysics of Running Couplings

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Abstract

In quantum field theories, couplings like charge, mass, and so on, vary—or ‘run’—with energy scale. This paper formulates and explores three metaphysically realist accounts of what running couplings are. Two accounts identify running couplings with properties; a third account identifies them with relations. As I argue, these accounts have different benefits and costs. For instance, they have different implications for issues—raised by Benitez, Fraser, Rivat, Ruetsche, Williams, and others—surrounding realism about effective theories. And more generally, these accounts help bring metaphysical theorizing into closer contact with contemporary particle physics.

1 Introduction

In the quantum field theories of the standard model, parameters like charge are not constants, fixed once and for all. Charge, mass, and other terms, vary across energy scales. The parameter corresponding to electron charge in quantum electrodynamics, for instance, has one value at a relatively low energy, another value at a higher energy, and various values at energies in between. Parameters

like these are often described as ‘running couplings’: the parameters – that is, the couplings¹ – run – that is, vary – with energy scale.

This paper is about the metaphysics of running couplings: I formulate, and explore the benefits and costs of, three accounts of what the physical contents of running couplings are. According to the ‘Property Account’, running couplings form hierarchies of monadic determinable properties, with corresponding determinates, across energy levels. According to the ‘Object Account’, instead of a hierarchy of properties, there is a hierarchy of particles across energy levels, and a running coupling is a single monadic determinable property—with corresponding determinates—which the particles in that hierarchy instantiate. According to the ‘Relation Account’, a running coupling is a binary determinable relation, with corresponding determinates, in which particles stand to energy levels.

It is important to develop metaphysical accounts of running couplings. Let me summarize two reasons why; more are discussed later. First, the metaphysics of running couplings is relevant to debates over realism about quantum field theories. According to attractive versions of realism (Fraser, 2020; Williams, 2019), quantum field theories are best understood as effective: they accurately describe physical phenomena only up to some particular, cutoff energy level Λ_c ; at higher energies, the theories cease to be reliable; they are no longer effective descriptions of the relevant physics.² Running couplings figure prominently in realist accounts of effective quantum field theories: renormalization group equations, which describe how couplings run, also characterize the scale-dependence that circumscribes the domain over which effective theories apply. So it is natural to think that running couplings are among the physical structures which comprise the physical contents of effective theories. And so it is important to give precise accounts of exactly what those physical structures, encoded in the mathematics of renormalization group equations,³ are: that is part of a larger project of specifying the physical contents of effective theories generally (Miller, 2023; Williams,

¹In this paper, I use ‘coupling’ to refer both to parameters associated with interaction terms in Lagrangians, and also to the mass parameter associated with free terms.

²For canonical references which helped facilitate this understanding of cutoffs in renormalized perturbation theory, see (Polchinski, 1984; K. G. Wilson & Kogut, 1974).

³Relatedly, by ‘running couplings’, I generally mean the physical contents—the worldly properties or relations, say—of the relevant formal expressions for scale-dependent mathematical parameters in renormalization group equations.

2023).

Second, exploring accounts of running couplings brings the metaphysics literature into closer contact with contemporary physical theorizing. The metaphysics of physics literature focuses on classical mechanics and non-relativistic quantum particle mechanics (Wallace, 2020). Given some of that literature’s goals, however, this focus is misdirected. Quantum field theories of the standard model—unlike either classical mechanics or non-relativistic quantum particle mechanics—represent arguably the first theories which offer any hope of providing a single unified model of all known physical interactions for length scales on, say, the surface of the Earth (Wallace, 2020, p. 4313). Insofar as the metaphysics of physics literature seeks guidance from our most unified theories of physics to date, the focus should be on quantum field theories. For that matter, insofar as metaphysics generally aspires to scientific naturalism, quantum field theories should be discussed much more, in the broader metaphysics literature, than they have been.⁴ The present paper helps with that by offering several metaphysical accounts of running couplings in quantum field theories.

Relatedly, since there are basically no accounts of the metaphysics of running couplings in the literature, my primary goal here is not to argue for any one account over the others – though on balance, the Relation Account strikes me as best. Rather, my primary goal is to chart several candidate accounts of what running couplings are, and discuss their relative benefits and costs. This will both (i) illustrate how different views of running couplings bear on debates over realism about effective theories (Ruetsche, 2011), and (ii) lay the groundwork for more physically realistic metaphysical theorizing (Baker, 2016), based on more up-to-date theories of particle physics. As I discuss later, this also has the virtue of illuminating an important region within the space of candidate answers to the difficult interpretive questions which renormalization raises. For example, these metaphysical models of running couplings—though in many ways speculative—are precise and detailed enough to facilitate fruitful comparisons with views which, for instance, take renormalization group transformations to yield physically equiv-

⁴For an explication of one illustrative conflict between metaphysical theories and quantum physics, see (Wilhelm, 2022). For an outline of the sort of realist program which this paper advances, focusing in particular on the proper framing of the question of what quantum field theories say about the world physically, see (Wilhelm, 2024).

alent parameterizations of the physics (Fraser, 2021; Rosaler & Harlander, 2019): views which would deny, to varying extents, one or more of the physical reifications of the renormalization formalism that I explore here. So ultimately, my project is both exploratory and conditional in nature: conditional on taking standard interpretations—what Ruetsche (2024) calls ‘folk’ interpretations—of scale-dependent running couplings seriously, this paper explores how best to model what, exactly, physical running couplings are.

To keep things concrete, I focus on the Lagrangian for quantum electrodynamics, and the scale-dependence of the couplings associated with the electron field. I focus on Lagrangian formalisms because they are perfectly respectable physical theories, suitable for philosophical interpretation (Wallace, 2006) and important for practicing physicists today. I focus on quantum electrodynamics because it is relatively simple, yet also highly accurate over the domain of energies and physical phenomena which it describes. Metaphysical accounts of running couplings for, say, the electroweak theory or quantum chromodynamics, are well worth pursuing; but that is a topic for another paper.

In Section 2, I provide additional motivation for the project of this paper, and I present the equations which describe running couplings in quantum electrodynamics. In Section 3, I formulate, explain, and evaluate the three accounts of the metaphysics of those running couplings.

A quick preparatory remark. In this paper, I do not argue for realism about running couplings over anti-realism. The goal here is more modest: to formulate and explore precise realist views about running couplings. What realist views about couplings are there, and what might motivate realists to adopt one such view over others? These are the questions which this paper addresses. Arguments over realism and anti-realism about running couplings are, of course, important. But at this stage of inquiry, such arguments would be premature. Absent precise realist views of what running couplings are, debating realism about couplings risks imprecision, and disconnection from the details of the theories involved. By proposing precise realist options, this paper lays the groundwork for productively engaging in those debates.

2 Preliminaries

2.1 Motivation

To further motivate developing accounts of what running couplings are, it is worth making more remarks about how such accounts are important for realism. First, take an issue raised by Fraser: to substantiate the sorts of claims that realists about effective theories make, “we need a precise characterization of the non-fundamental descriptive content of QFTs” (2020, p. 289). A view which only characterizes the ontologies of effective theories in terms of posits which anti-realists accept—perhaps, for example, correlation functions understood as only encoding information about the outcomes of scattering events (Fraser, 2020, pp. 289-290; Ruetsche, 2020, p. 306)—is insufficient. Realists need to propose accounts of physical objects, properties, structures, and so on, beyond those compatible with anti-realism. Accounts of the metaphysics of running couplings contribute to that.

Second, running couplings are central to contemporary understandings of quantum field theories. The renormalization group equations, which describe how couplings run, also convey how to incorporate high-energy parts of a given theory’s path integral into a low-energy theory through modifications of couplings (Williams, 2022, p. 307). Much current evidence for quantum field theories draws on, and confirms, running couplings: see the experimental results presented in (Bethke et al., 2016; Denig & Salmè, 2013), for instance; or see the comprehensive reports of couplings’ scale-dependence in (Navas et al., 2024). In many ways, the phenomenon of running couplings is central to how quantum field theories work. So it is important to explore accounts of what couplings are.

Third, metaphysical accounts of running couplings might—depending on their details—support realism against specific anti-realist arguments. Take an argument due to Ruetsche (2024), who notes that running couplings raise a problem for a seemingly reasonable ontology to which realists about effective theories may commit. The ontology consists of particles and their nomically significant traits; the particles and traits at issue are encoded in the field terms, and couplings,

of an effective Lagrangian (Ruetsche, 2024, p. 19). Many effective Lagrangians associated with the bare Lagrangian of quantum electrodynamics, for instance, describe two species of particles—electrons and photons—which are essentially individuated by the nomically significant traits that the electron charge coupling e and the electron mass coupling m encode.⁵ The problem for this seemingly reasonable ontology, Ruetsche argues, is that those couplings cannot be essential to—and so cannot individuate—the particles at issue, since those couplings run: they change with energy scale, so they cannot be essential to the particles.

One lesson to draw from Ruetsche’s argument is encapsulated in the criterion below.

The Individuation Desiderata

Metaphysical accounts of running couplings should support a plausible theory of what individuates any posited particles.

At least two accounts of couplings, proposed here, satisfy the Individuation Desiderata; the third may or may not, depending on how best to interpret certain details of Ruetsche’s argument.

Fourth, there are very few satisfactory candidate ontologies for quantum field theories (Ruetsche, 2024, p. 19; Williams, 2023, p. 71). The metaphysical accounts of running couplings, proposed here, help address this: they provide candidate ontologies of couplings which are satisfying along different dimensions. Moreover, clarity on the metaphysics of running couplings can provide insight into the dynamical behaviors which particles and fields exhibit. For example, Benitez posits an ontology of effective particles, and suggests considering these particles as bundles of interaction charges (2023, p. 20).⁶ The present paper provides detailed metaphysical accounts of exactly what the elements of that bundle might be.⁷ More generally, metaphysical accounts of running couplings help address the

⁵Some effective Lagrangians associated with this bare Lagrangian of quantum electrodynamics, like the Euler-Heisenberg Lagrangian, integrate out the electron field entirely (Weinberg, 1995a, p. 32). Other effective Lagrangians retain both the electron field and the photon field, by integrating out phenomena at scales far above the electron mass coupling.

⁶For more bundle-theoretic approaches to the ontology of quantum field theories, see (Kuhlmann, 2010; Wayne, 2008).

⁷For more discussion of particle ontologies in quantum field theories, see (Baker, 2016, p. 5).

“interesting and challenging” (Williams, 2023, p. 27) question of exactly what role, in quantum field theories, is played by nonfundamental particles: the role concerns the running of their couplings. And more generally still, progress on the metaphysics of running couplings contributes to the overall goal of providing a complete ontology for quantum field theories.

2.2 Running Charge and Running Mass

In this section, I present renormalization group equations for running couplings. They are derived from the bare Lagrangian for quantum electrodynamics. One equation describes how a renormalized charge parameter varies with the renormalization scale Λ , and the other describes how a renormalized mass parameter varies with Λ . As is standard, Λ can be chosen to track a physical, external scale suitable for the scattering process at hand.

Two points of clarification. First, there are different methods for deriving equations which describe running couplings in quantum electrodynamics. Derivations based on Wilsonian methods have the advantage of being more physically transparent, especially with respect to calculations of the partition function: one introduces a finite explicit cutoff and then changes the parameters of the relevant Lagrangian in order to ensure the cutoff-independence of the physical quantities. But when dealing with renormalizable theories like quantum electrodynamics, the complexities of integrating out slices in momentum space—and introducing an infinite number of interaction types—are inconvenient. Derivations based on ordinary renormalized perturbation theory, like those presented here, are technically easier (Peskin & Schroeder, 1995, p. 406; Weinberg, 1995a, p. 112).

Second, different derivations yield slightly different approximations of how running couplings vary with energy scale, and there are different physical and metaphysical interpretations of that. For instance, some derivations truncate the relevant expansions at leading order – like in the derivation presented here – while others truncate at higher orders – see (Weinberg, 1995a, pp. 119-127).

See also the discussion of particles as regime-dependent excitations in (Wallace, 2020, p. 4314). All of the accounts of running couplings, proposed here, are compatible with understanding particles along these lines.

And there are different interpretations of what this means. According to the more coarse-grained interpretation which I prefer, each approximation is merely a mathematical estimate of the true facts about how charge and mass quantities run with energy scale. It is commonplace, in all areas of science, to calculate quantities using different approximations. So long as those different approximations agree with one another to within some error margin, this is compatible with taking the different approximations to be different mathematical estimates of the true facts about the quantities at issue. Other interpretations are possible, however. For instance, one might take the many different approximations of running charge, and mass, to represent many different collections of physical charge and mass quantities, each of which run in slightly different ways. Or one might take the many different approximations of running charge, and mass, to indicate some sort of agent-relativity, or calculational-scheme-relativity, of the physical quantities at issue. These other interpretations are compatible with the accounts presented later: they just require rewording those accounts, to explicitly mention the many different collections of quantities or the calculational-relativity of quantities.⁸

Now to present the renormalization group equations for running charge and running mass. The bare Lagrangian for quantum electrodynamics is

$$\mathcal{L} = -\frac{1}{4}F_{\mu\nu}^2 + \bar{\psi}^0(i\cancel{\partial} - m^0)\psi^0 - e^0\bar{\psi}^0\gamma^\mu A_\mu^0\psi^0 \quad (1)$$

where ψ^0 is the bare—that is, unrenormalized—field of the electron, A_μ^0 is the bare field of the photon, $F_{\mu\nu} = \partial_\mu A_\nu^0 - \partial_\nu A_\mu^0$ is the bare electromagnetic field-strength tensor, the Dirac matrices γ^μ facilitate descriptions of spin $\frac{1}{2}$ particles, ∂^μ and $\cancel{\partial} = \gamma^\mu\partial_\mu$ are derivative operators on relativistic spacetime, $e^0 = -|e^0|$ is the electron’s bare charge, m^0 is the electron’s bare mass, and the Einstein summation notation is being employed (Peskin & Schroeder, 1995, p. 78). The

⁸Arguably, this coarse-grained view is standard in science generally and in particle physics specifically. Take Schwartz, who observes that whereas one derivation of the renormalization group equation for charge uses logarithms (2014, p. 309-314), a later derivation of approximately that same equation uses counterterms instead (2014, p. 425). Schwartz does not conclude, from this, that there are many marginally different sorts of running charge which these marginally different approximations describe, or that facts about running charge are logarithm-relative and counterterm-relative. Instead, Schwartz notes that given reasonable assumptions, these approximations describe the same running charge.

renormalized electron field, photon field, mass, and charge, are defined as follows:

$$\psi = \frac{1}{\sqrt{Z_2}}\psi^0 \quad A_\mu = \frac{1}{\sqrt{Z_3}}A_\mu^0 \quad m_R = \frac{1}{Z_m}m^0 \quad e_R = \frac{1}{Z_e}\Lambda^{\frac{d-4}{2}}e^0 \quad (2)$$

where the constants Z_2 , Z_3 , Z_m , and Z_e are electron field strength, photon field strength, mass, and charge renormalizations respectively, Λ is a particular energy level that plays the role of a renormalization scale, and $d = 4 - \epsilon$ represents the number of spacetime dimensions given a small deviation parameter ϵ (Schwartz, 2014, p. 424).⁹ Substituting these into (1) yields the renormalized Lagrangian below.

$$\mathcal{L} = -\frac{1}{4}Z_3F_{\mu\nu}^2 + iZ_2\bar{\psi}\not{\partial}\psi - m_RZ_2Z_m\bar{\psi}\psi - \Lambda^{\frac{4-d}{2}}e_RZ_eZ_2\sqrt{Z_3}\bar{\psi}\gamma^\mu A_\mu\psi$$

Because there is scale-dependence in the renormalized Lagrangian but not the bare Lagrangian, the following holds.

$$\Lambda \frac{d}{d\Lambda} e^0 = 0$$

Writing e^0 in terms of $e_R = e_R(\Lambda)$ as in (2), differentiating, truncating, and taking ϵ to 0, yields the following (Schwartz, 2014, pp. 424-426).

$$\Lambda \frac{d}{d\Lambda} e_R(\Lambda) = \frac{e_R^3}{12\pi^2} \quad (\text{Ch})$$

This is the leading-order quantum electrodynamic renormalization group equation for charge. It describes how charge varies with energy level. The explicit solution at leading order is

$$\frac{e_R^2(\Lambda)}{4\pi} = -\frac{3\pi}{2 \ln \left(\frac{\Lambda}{\Lambda_{QED}} \right)}$$

where Λ_{QED} , the location of the Landau pole, is an integration constant fixed by the boundary conditions (Schwartz, 2014, p. 425).

Equation (Ch) specifies the quantitative value of the renormalized charge

⁹These renormalizations are chosen so that the terms in the renormalized QED Lagrangian have certain desirable properties. Z_m is chosen, for instance, so that m_R is the location of the pole in the exact electron 2-point function (Schwartz, 2014, p. 340).

coupling $e_R(\Lambda)$ as a function of the energy scale Λ . Vary the energy scale, and the renormalized charge varies accordingly, in the manner that (Ch) specifies. That is the sense in which charge runs in quantum electrodynamics.

An argument similar to the one just given can be used to derive the following renormalization group equation for mass (Schwartz, 2014, pp. 425-426).

$$\frac{\Lambda}{m_R(\Lambda)} \frac{d}{d\Lambda} m_R(\Lambda) = -\frac{3e_R^2(\Lambda)}{8\pi^2} \quad (\text{Ma})$$

This equation specifies the quantitative value of the renormalized mass coupling $m_R(\Lambda)$ as a function of Λ : it specifies how mass runs in quantum electrodynamics.

Note that in general, the choice of Λ in (2) is not fixed by the particular physical process being analyzed. Λ can be an arbitrarily chosen renormalization scale,¹⁰ introduced in order to define the relevant renormalization parameters. Real physical processes are characterized by an external momentum scale Q , and in perturbative calculations, observables are often expressed in terms of both the physical scale parameter Q and the renormalization scale parameter Λ .

The scale Λ can be chosen to approximate the scale Q (Schwartz, 2014, p. 313).¹¹ And the scale-dependence of observables can be represented by evaluating the renormalized couplings for that choice of Λ . This makes the charge and mass couplings, expressed as functions of Λ , scheme-dependent representations of the scale-dependent effective couplings – consequently, $e_R(\Lambda)$ and $m_R(\Lambda)$ are mathematical proxies for the effective charge and effective mass that vary along the external physical scale.¹² So throughout this paper, as I continue to use Λ for the renormalization scale parameter, understand this Λ as chosen to track the physical scale appropriate for the physical process at issue.¹³

¹⁰ Λ is, in this sense, a “fake parameter” (Srednicki, 2007, p. 174). As I explain later, Λ can be chosen to approximately coincide with an external scale which is physically significant insofar as it tracks the relevant physical processes.

¹¹At any finite perturbative order, predictions retain residual Λ -dependence, since higher-order terms are truncated from the calculations. It is standard practice to estimate the uncertainty associated with that truncation by varying Λ within a reasonable range around Q , treating the resulting spread as an estimate of truncation error. The present discussion therefore takes the physical, realist content of running couplings to be the stable structures that persist across reasonable scale-setting choices and their associated uncertainty estimates.

¹²For an example of the mathematical equations relating these proxies to the effective couplings, see (Schwartz, 2014, pp. 313-314).

¹³As is often observed, this choice of renormalization scale Λ can also have the virtue of

An objection related to this is worth discussing. Different renormalization schemes occasionally generate different equations for running couplings. The renormalization group equations (Ch) and (Ma) were calculated using the modified minimal subtraction scheme. But there are other schemes which produce other renormalization group equations; the beta functions, describing how electron charge varies with energy scale, depend on the scheme at issue. So any purportedly metaphysically realist account of renormalized charges and masses in quantum electrodynamics is problematically scheme-dependent. And so any such account cannot be realist after all.

There are several reasonable responses to this. The first response: at one-loop, many different renormalization schemes yield the same equations; the same beta function for electron charge, for instance. Calculations using the momentum subtraction scheme yield (Ch) too, as do calculations using the Wilsonian scheme and the on-shell subtraction scheme (Srednicki, 2007, pp. 162-175). So renormalized charge and mass are not problematically scheme-dependent. And so there is no problem, here, for realist metaphysical theorizing based on (Ch) and (Ma).

The second response: more generally, the scheme-dependence of various calculations need not imply any anti-realism about the quantities calculated. It is true that at two-loop and beyond, different renormalization schemes produce different equations for running couplings. But that is compatible with realism about running charge and mass. For plausibly, all these different renormalization schemes provide different approximations, to varying degrees of accuracy, of the maximally detailed beta functions which incorporate all relevant couplings in the standard model; and consequently, provide the most accurate description of those couplings' scaling behaviors.¹⁴ And those behaviors, in particular the

minimizing large logarithmic contributions of higher-order terms in the calculations (Trócsányi, 2015, p. 24).

¹⁴In addition to electromagnetic interactions, hadronic and electroweak interactions also contribute to the electron charge at low energies: the calculations of running couplings are, in this sense, extremely holistic. For calculation of the quantum electrodynamic contribution up to tenth order—accounting for, among other things, hadronic contributions, estimates of the two-loop electroweak effect, and the electron's self-energy—see (Aoyama et al., 2012). Arguably, these precision tests amount to “closing the loop” (Smith, 2014) by using quantum electrodynamical principles to uncover slight discrepancies between measured and calculated values for various parameters; those discrepancies can then guide improvements to the background quantum electrodynamical model – all while maintaining the general correctness of the framework

commonalities shared by all schemes, are what to be realists about.¹⁵

One final comment on the scope and limits of this paper.¹⁶ There are many substantial interpretive questions—more than I can cover here—about what ‘runs’ with energy scale in quantum field theories. For instance, one might wonder whether the scale-dependent variation in mathematical quantities like $e_R(\Lambda)$ and $m_R(\Lambda)$ really does represent a ‘running’ of corresponding physical structures in the world. Perhaps for any choice of renormalization scale Λ , $e_R(\Lambda)$ and $m_R(\Lambda)$ are best interpreted as non-invariant, coordinate-like descriptions of quantum electrodynamics’ scale dependence. If so, then reifying $e_R(\Lambda)$ and $m_R(\Lambda)$ amounts to problematically ontologizing a particular formal representation of running couplings – translating that formalism into ill-fitting, object-property metaphysics.

The present paper does not argue against this alternative approach to interpretive questions about running couplings. Instead, the present paper should be read as an investigation of how best to model reifications of $e_R(\Lambda)$ and $m_R(\Lambda)$, conditional on the assumption that $e_R(\Lambda)$ and $m_R(\Lambda)$ are not purely formal, coordinate-like descriptions of scale dependence. Folk interpretations of quantum field theories often describe there being different values of electron charge and mass at different scales (Ruetsche, 2024). It is worth exploring how such talk might be modeled metaphysically – and that is what I do here. Among other things, this has the virtue that the resulting metaphysical models—the Property Account, the Object Account, and the Relation Account—are sufficiently precise and cleanly defined to facilitate comparisons with alternative approaches to interpretive questions like those mentioned above. For instance, these metaphysical models can be compared to the view that perturbative renormalization schemes

(Koberinski & Smeenk, 2020). For more on the different formal techniques that comprise renormalization group methods, and the epistemological issues which those techniques generate, see (Koberinski & Fraser, 2023).

¹⁵Thanks to an anonymous reviewer for this point. It is also worth flagging that there are other interesting options for realists to explore. According to one option, there are many different sorts of running charge and running mass, one for each scheme which generates a different renormalization group equation – and the three metaphysically realist accounts of running couplings, proposed here, can be adapted to each scheme. According to another option, just one scheme is privileged, in that it alone—among all schemes—yields the renormalization group equations which most accurately describe the metaphysically realist facts about running charge and mass. These alternatives strike me as less plausible than the option described in the main text; but it is worth exploring them in future work.

¹⁶Many thanks to an anonymous reviewer for discussion of these points.

are physically equivalent notational variants (Fraser, 2021, p. 126), or to the view that even within the Wilsonian approach, renormalization group transformations are best understood as physically equivalent parameterizations of the same physical quantities (Rosaler & Harlander, 2019, p. 125).

So think of this paper as an exercise in physics-inspired speculative metaphysical theorizing. Renormalization raises deep interpretive difficulties concerning which elements of quantum formalisms are physically significant and which are not. In order to make some progress on at least some of these difficulties, I develop and explore three metaphysical models, situated within a collection of reasonable background realist assumptions about the renormalization formalism, inspired by the thought that charge and mass really do differ across different scales. And so the proper reading of this paper is not that one of the three metaphysical models must be right; it would be harmful to the philosophical literature if one of these models became ossified as the only decent option for realists about quantum electrodynamics. Rather, the proper reading is that there are three precise metaphysical models which do a reasonably good job of respecting the folk interpretation of renormalization, each of which comes with its own benefits and costs – and each of which is detailed enough to facilitate serviceable comparisons with other interpretations of renormalization group transformations in future work.

3 Three Metaphysical Accounts

This section explores metaphysical accounts of running charge, and mass, which specify the physical contents of equations (Ch) and (Ma). In Section 3.1, I present the Property Account, which indexes properties to energy levels. In Section 3.2, I present the Object Account, which indexes objects to energy levels. In Section 3.3, I present the Relation Account, which posits relations.

Two clarificatory remarks. First, though the accounts focus on running charges and masses for electrons, nothing hangs on that choice of particle. The same basic claims apply to running charges and masses for all charged leptons well-described by quantum electrodynamics.

Second, in what follows, I use simple formalisms to regiment claims about electrons’ charges and masses which the different accounts can be used to make. This is not pedantry. The formal regimentations make important differences between the accounts salient.

3.1 The Property Account

According to the Property Account of running couplings, charges and masses are monadic properties which electrons exemplify. But unlike charges and masses in classical mechanics, the Property Account posits a continuous infinity of charge properties and a continuous infinity of mass properties: one such charge, and one such mass, for each energy level Λ below a specified cutoff Λ_c that defines the regime in which quantum electrodynamics applies. In addition, the Property Account posits a collection of specific charge quantities and a collection of specific mass quantities – defined using (Ch) and (Ma) respectively – for each Λ .

Here are the details. For each energy level Λ below Λ_c , there is a monadic determinable charge property C_Λ with many monadic determinates c_Λ^r . The subscript ‘ Λ ’ indicates that there is one charge determinable, and one corresponding collection of charge determinates, for each Λ below the cutoff. The superscript ‘ r ’ indicates that there is a charge determinate c_Λ^r , of the given C_Λ , for every scalar quantity of charge r which a particle may have. Likewise for mass: for each energy level Λ below Λ_c , there is a monadic determinable mass property M_Λ with many monadic determinates m_Λ^s , and there is one such determinate for each allowed scalar quantity of mass. Each C_Λ is the property of being charged-at- Λ , each M_Λ is the property of being massive-at- Λ , each c_Λ^r is the property of having r -amount-of-charge-at- Λ , and each m_Λ^s is the property of having s -amount-of-mass-at- Λ . That is the Property Account of running couplings.

A subtle but crucial point: the use of hyphens in the descriptions “ C_Λ is the property of being charged-at- Λ ” and “ c_Λ^r is the property of having r -amount-of-charge-at- Λ ” is important. English phrases like “The electron has r amount of charge at Λ ” or “The electron’s charge varies with energy scale”—used in various arguments in the literature; see the examples in Section 2.1—are ambiguous in

ways which sometimes matter for philosophical accounts of effective theories. The three metaphysical accounts of running couplings provide different precisifications of those English phrases, and in so doing, resolve the ambiguity. The Relation Account, discussed later, interprets those phrases by positing a two-place relation between the electron and the energy level Λ . The Property Account is quite different: for every single Λ below Λ_c , it posits a one-place charged-at- Λ property.

To illustrate these features of the Property Account, note that for each energy level Λ below the cutoff Λ_c , electron e has both determinables C_Λ and M_Λ . Put more formally: for each Λ below Λ_c , $C_\Lambda(e)$ and $M_\Lambda(e)$. In addition, for each Λ below Λ_c , e has the determinate c_Λ^r where r is the quantity given by equation (Ch) corresponding to Λ , and e has the determinate m_Λ^s where s is the quantity given by equation (Ma) corresponding to Λ . More formally, for the specific r and s which (Ch) and (Ma) respectively output when given input Λ , $c_\Lambda^r(e)$ and $m_\Lambda^s(e)$; note that $c_\Lambda^r(e)$ is the Property Account's regimentation of the ambiguous English phrase "The electron has r amount of charge at Λ " discussed earlier. For any two different energy levels Λ and Λ' below Λ_c , the determinate c_Λ^r property, of e , differs from the determinate $c_{\Lambda'}^{r'}$ property of e – where r' is the quantity given by equation (Ch) corresponding to Λ' – and the determinate m_Λ^s property of e differs from the determinate $m_{\Lambda'}^{s'}$ property of e – where s' is the quantity given by equation (Ma) corresponding to Λ' . But there is nothing problematic about this. For c_Λ^r and $c_{\Lambda'}^{r'}$ are determinates of the numerically distinct determinables C_Λ and $C_{\Lambda'}$ – and m_Λ^s and $m_{\Lambda'}^{s'}$ are determinates of the numerically distinct determinables M_Λ and $M_{\Lambda'}$.

It is worth explaining this in a bit more detail: to do so, I focus on the case for charge; the case for mass is analogous. The following is a standard principle of how determinables and determinates relate.

Preclusion

For all x , all determinables D , and all distinct determinates d_1 and d_2 of D , if x has d_1 then x does not have d_2 (Elder, 2023; J. Wilson, 2023).

In other words, different determinates of the same determinable are incompatible; having one precludes having the other. Preclusion, along with the Property Account, implies that for every Λ below Λ_c , (i) for the r given by (Ch), e has c_Λ^r , (ii)

c_Λ^r is a determinate of the determinable C_Λ , and therefore (iii) for every r' distinct from r , e does not have the determinate $c_\Lambda^{r'}$ of C_Λ . Having the determinate c_Λ^r of C_Λ , that is, precludes having any other determinate $c_\Lambda^{r'}$ of C_Λ . But notice that by the Property Account, for different energy levels Λ and Λ' , the determinables C_Λ and $C_{\Lambda'}$ are distinct: charge-at- Λ is one property; charge-at- Λ' another. In addition, by the Property Account, the associated families of determinates are completely distinct too: so in particular, r -amount-of-charge-at- Λ – that is, c_Λ^r – is not a determinate of $C_{\Lambda'}$, and r' -amount-of-charge-at- Λ' – that is, $c_{\Lambda'}^{r'}$ – is not a determinate of C_Λ . So the Property Account implies that for distinct Λ and Λ' below Λ_c , (i) for the r given by (Ch), e has c_Λ^r , (ii) c_Λ^r is a determinate of the determinable C_Λ but not the determinable $C_{\Lambda'}$, (iii) for the r' given by (Ch), e has $c_{\Lambda'}^{r'}$, (iv) $c_{\Lambda'}^{r'}$ is a determinate of the determinable $C_{\Lambda'}$ but not the determinable C_Λ , and (v) none of this contradicts Preclusion.

Now to review some benefits and costs of the Property Account. One argument in the account's favor: it addresses the issue raised by Fraser from Section 2.1. As Fraser suggests, realism about quantum field theories requires a precise characterization of those theories' non-fundamental, descriptive content – and that characterization needs to go beyond posits compatible with anti-realism. The Property Account does this, by providing a precise characterization of the non-fundamental, descriptive content of running couplings in quantum electrodynamics: for each Λ , there is one monadic determinable charge-at- Λ property, one monadic determinable mass-at- Λ property, a collection of monadic determinate properties of having r -amount-of-charge-at- Λ – exactly one of which, for the r given by (Ch) corresponding to Λ , is what electrons actually exemplify – and a collection of monadic determinate properties of having s -amount-of-mass-at- Λ – exactly one of which, for the s given by (Ma) corresponding to Λ , is what electrons actually exemplify.

The properties posited by the Property Account are robustly realist and unfriendly to anti-realism. There are charge-at- Λ properties, mass-at- Λ properties, r -amount-of-charge-at- Λ properties, and s -amount-of-mass-at- Λ properties, for Λ beyond the energy scales which experiments can test. The Property Account posits monadic determinable charge and mass properties, and monadic determinate charge and mass properties, up to an appropriate cutoff for quantum

electrodynamics understood as an effective theory.

Note that anti-realists could endorse a modified version of the Property Account.¹⁷ According to this modified version, there are charge-at- Λ properties, mass-at- Λ properties, r -amount-of-charge-at- Λ properties, and r -amount-of-mass-at- Λ properties, for Λ only up to an energy scale Λ^* well below the cutoff for quantum electrodynamics understood as an effective theory. This could be further supported by constructing the beta functions for charge and mass from measurements of sufficiently low-energy reference scales. Such an anti-realist modification of the Property Account might be somewhat arbitrary, as it makes an arbitrary choice about exactly which low-energy reference scale to use. But this anti-realist modification is at least a possibility for those uncomfortable with the realist posits of the Property Account presented here.

One might worry that the Property Account contradicts the realist strategy of ontologically committing to those elements of effective theories that are “invariant across independent and distinct choices about how to model the physics at the short distances where the theory is empirically inapplicable” (Williams, 2019, p. 220). For according to the Property Account, one might claim, charge and mass vary with energy scale; charge and mass alternate depending on the scale being examined. And so, one might claim, the Property Account implies that charge and mass are not invariant across distinct modeling choices; charge and mass are not robust in the sense that realism, about those properties, requires.

In two ways, this objection is mistaken. First, the Property Account does not imply that charge and mass vary, in the objection’s sense, with energy scale. It is not as if there is one charge property, and whether or not e has that one property varies with Λ . The objection describes the Property Account using phrases which are ambiguous in the ways that, as explained earlier, “The electron’s charge varies with energy scale” is ambiguous. The Property Account resolves the ambiguities by positing a precise ontology of properties. And whether or not a given particle instantiates a determinable like c_Λ^r is, indeed, invariant across energy scales: ‘varying’ the energy scale, or ‘examining’ a different energy scale—whatever ‘varying’ or ‘examining’ might mean—does not change the fact that the electron has the property c_Λ^r .

¹⁷Thanks to an anonymous reviewer here.

Second, the ontological posits of the Property Account satisfy a standard realist criterion of robust invariance. The robustness criterion asserts that the physically significant elements of theories, worth adopting as ontological posits, include those which are accessible—derivable, identifiable, measurable, and so on—through many independent methods, including methods employing different background assumptions or models or elementary principles (Wimsatt, 2007, pp. 44-45).¹⁸ Running charge and mass are accessible in this sense. As discussed in Section 2.2, different methods of deriving equations for running couplings, Wilsonian or renormalized perturbative, converge on the same results. Convergent measurements suggest the physical reality of running couplings (Navas et al., 2024). And different models of high energy phenomena converge, under renormalization group flow, to the same running coupling behavior (Williams, 2019, pp. 219-220). The Property Account adopts, as ontological posits, properties which respect all this. So it is false to claim, as the objection does, that the Property Account fails to respect the robust invariance of charge and mass.

Another benefit of the Property Account: it satisfies the Individuation Desiderata, and so can be used to defend realism against Ruetsche’s argument. For consider the following theory, which the Property Account supports, of what individuates electrons: the essential attributes of electrons are, for each Λ below Λ_c , the determinable properties C_Λ and M_Λ , and the corresponding determinates c_Λ^r and m_Λ^s for r and s given by (Ch) and (Ma); part of what it is to be an electron, in other words, is to instantiate C_Λ , M_Λ , c_Λ^r , and m_Λ^s . Other particles fail to instantiate at least some of those properties. Neutrinos, for instance, have mass as described by certain extensions of the original standard model, and that mass runs: so neutrinos have determinable mass properties like M_Λ . But neutrino mass runs according to an equation which is different from (Ma), so for a given Λ , neutrinos have a determinate mass property $m_\Lambda^{s^*}$ for a specific s^* which differs from the specific s given by (Ma) corresponding to Λ – and so neutrinos do not have the determinate mass property m_Λ^s which electrons have.¹⁹

Yet another benefit of the Property Account: it supports the bundle theory

¹⁸For more on the importance of adopting criterion like this, see (Stanford, 2006, p. 169).

¹⁹Even other charged leptons, like muons and taus, have slightly different renormalization group equations (Navas et al., 2024)—qualitatively similar to (Ch) and (Ma) but nevertheless distinct—because of differences in their masses (Weinberg, 1995a, p. 530; 1995b, pp. 309-310).

of effective particles discussed by Benitez (2023). For electrons could be identified with bundles of, among other properties, the determinables C_Λ and M_Λ and their corresponding determinates c_Λ^r and m_Λ^s . What it is to be an electron, on this view, is to be a collection of properties which includes, for each energy level Λ , the determinable charge-at- Λ and mass-at- Λ properties, and the corresponding determinate properties of having r -amount-of-charge-at- Λ and s -amount-of-mass-at- Λ . This attractively reduces particles to properties.

The Property Account is consistent with other accounts of particles, however. It can be combined with the view that particles are thin particulars (Sider, 2006) which instantiate a continuous infinity of different determinable charge, determinable mass, determinate charge, and determinate mass properties. In that case, what connects the distinct properties is the primitive fact that one and the same thin particular instantiates them. And given that particles are thin particulars, they are individuated in the manner that thin particulars typically are: by the different properties which they instantiate, which the renormalization equations characterize; and also perhaps by primitive identity facts about which thin particulars are distinct from which.

One more significant benefit of the Property Account is worth mentioning. The Property Account does not require abandoning a standard semantic tenet of scientific realism: scientific theories, and the sorts of descriptions which they provide, are true or false simpliciter (Psillos, 1999, p. xvii). As Rivat observes, certain realist descriptions of continuum fields may need to be evaluated, semantically, as “true or false relative to a specific range of physical scales” (2021, p. 12133) – in other words, realists must index truth itself to scales of energy. For the properties which individuate those fields are specified in arbitrarily small regions where the background effective theories break down: and this provides reason for thinking that effective theories’ descriptions of continuum fields are not even close to being approximately accurate (Rivat, 2021, p. 12123).

The Property Account, and in particular the realist commitment to C_Λ , M_Λ , and all the c_Λ^r and m_Λ^s , avoids this. Equations (Ch) and (Ma) are accurate representations of the physical facts about running charge and mass. And most importantly, the Property Account captures the scale-relativity of those facts by indexing determinable charge, determinate charge, determinable mass, and de-

terminate mass, to energy levels. So there is no need to index truth itself. The indexing shows up in the continuum of charge and mass properties, structured into a hierarchy described by renormalization group equations, which the Property Account provides.

Though the Property Account has lots of benefits, it has some costs too. First, it might seem unparsimonious, since it posits infinitely many different determinable charge properties and infinitely many different determinable mass properties – and correspondingly, infinitely many different collections of infinitely many determinate charge properties and determinate mass properties. One might want a metaphysical account of running couplings which is less ontologically profligate.

Second, the Property Account does not provide a theory of exactly what unites all the different determinable charge properties and determinable mass properties. There are nomological expressions which unify them: equations (Ch) and (Ma). But absent an endorsement of something like nomic essentialism, this cannot provide an account of, for instance, what makes C_Λ and $C_{\Lambda'}$ —for distinct Λ and Λ' —count as similar. In virtue of what do they resemble each other? In virtue of what are they aptly called by the same name ‘charge determinables’? The Property Account, by being silent on these questions, seems committed to problematically brute, unexplained similarity among its posits.

These are not decisive reasons to reject the Property Account. But they are costs. So it is worth exploring alternative accounts of the metaphysics of running couplings, and seeing how the Property Account compares.

3.2 The Object Account

In this section, I present the Object Account of running couplings. According to the Object Account, charge and mass are monadic properties which electrons exemplify. Unlike the Property Account, however, there is just one determinable charge property and just one determinable mass property; and correspondingly, just one collection of determinate charges and determinate masses for each. But in addition, the Object Account posits a continuous infinity of electrons: one for each energy level below the cutoff Λ_c .

Here are the details. There is a single monadic determinable charge property C with monadic determinates c^r , and there is a single monadic determinable mass property M with monadic determinates m^s . As before, the superscripts ‘ r ’ and ‘ s ’ indicate that there is a charge determinate c^r of C , and a mass determinate m^s of M , for every single scalar quantity of charge r – and scalar quantity of mass s – which an object may have. Note, however, that there are no energy level subscripts on C , M , the c^r , or the m^s . Instead, for each Λ below Λ_c , there is a particle e_Λ ; call it a ‘ Λ -electron’. And for distinct Λ and Λ' below Λ_c , e_Λ is different from $e_{\Lambda'}$. That is the Object Account of running couplings.

Here is how the Object Account can be used to describe facts about running charge and running mass. For each Λ below Λ_c , the Λ -electron e_Λ has both determinables C and M . Put more formally: for each Λ below Λ_c , $C(e_\Lambda)$ and $M(e_\Lambda)$. In addition, for each Λ below Λ_c , e_Λ has the determinate c^r where r is the quantity given by (Ch) corresponding to Λ , and e_Λ has the determinate m^s where s is the quantity given by (Ma) corresponding to Λ . More formally, $c^r(e_\Lambda)$ and $m^s(e_\Lambda)$; note that $c^r(e_\Lambda)$ is the Object Account’s regimentation of the ambiguous English phrase “The electron has r amount of charge at Λ ” discussed earlier. For any two distinct Λ and Λ' below Λ_c , the determinate c^r charge property of e_Λ differs from the determinate $c^{r'}$ charge property of $e_{\Lambda'}$ – where r' is the quantity given by (Ch) corresponding to Λ' – and the determinate m^s property of e_Λ differs from the determinate $m^{s'}$ property of $e_{\Lambda'}$ – where s' is the quantity given by (Ma) corresponding to Λ' . But that is fine: e_Λ and $e_{\Lambda'}$ are different objects, so they can have different determinates of the same determinables. And so there is no conflict with Preclusion.

The Object Account has many of the benefits which the Property Account has. For instance, the Object Account addresses the issue raised by Fraser. Given the Object Account, for each Λ below the cutoff, there is a Λ -electron e_Λ which has both monadic determinable properties C and M , as well as the corresponding monadic determinate properties c^r and m^s from (Ch) and (Ma) respectively. These objects and properties are all robustly realist and unfriendly to anti-realism, since they concern energies far above the levels at which experiments can probe; though as in the case of the Property Account, one might adopt an anti-realist modification of the Object Account which only posits a Λ -electron e_Λ for Λ up to

a suitably low-energy, empirically accessible energy scale. In addition, the objects and properties of the Object Account are invariant across independent choices about how to model the physics at extremely high energies: the properties C and M , and the facts about which Λ -electrons instantiate which determinate properties c^r and m^s , exhibit that invariance. So the Object Account is friendly to both the letter and the spirit of realism about effective theories.

In addition, the Object Account supports Benitez's bundle theory of effective particles. Each e_Λ may be identified with bundles of, among other properties, the determinables C and M along with the relevant determinates c^r and m^s – note that given equations (Ch) and (Ma), it follows that (i) the determinates c^r and m^s in the bundle identified with e_Λ , differ from (ii) the determinates $c^{r'}$ and $m^{s'}$ in the bundle identified with $e_{\Lambda'}$. What it is to be a Λ -electron, on this view, is to be a bundle of properties which includes those determinables and determinates.

Note that this combination of the Object Account and the bundle theory of particles differs from the combination of the Property Account and the bundle theory of particles. The former combination posits a continuous infinity of particles: there is a Λ -electron for each collection of the four properties C , M , c^r , and m^s , where r and s are the scalars appropriate for Λ . The latter combination posits a single particle: there is just one electron, for the collection consisting of a continuous infinity of properties C_Λ , M_Λ , c_Λ^r , and m_Λ^s , one for every Λ . So the combination of the Object Account and the bundle theory does not collapse into the combination of the Property Account and the bundle theory: they identify particles with different bundles.

The Object Account does not require abandoning the standard realist tenet that scientific theories are true or false simpliciter. For the Object Account captures the scale-relativity of facts about running couplings by indexing electrons to energy levels. So there is no need to index truth itself.

Nevertheless, the Object Account has costs. Perhaps the main potential cost concerns whether it satisfies the Individuation Desiderata – and so, whether it can be used to defend realism against Ruetsche's argument. The most plausible view of particles' essences and particle individuation, which the Object Account supports, is as follows: for each Λ below Λ_c , the determinables C and M are essential attributes of e_Λ , as are the corresponding determinates c^r and m^s . The problem:

this seems to individuate particles too finely, by drawing more distinctions than the Individuation Desiderata recognizes. For the Object Account implies that given any distinct Λ and Λ' below Λ_c , the Λ -electron e_Λ is distinct from the Λ' -electron $e_{\Lambda'}$. So this view does not just individuate the electron particle species from neutrino particle species, quark particle species, and so on. Rather, this view individuates different particle species among electrons themselves: it distinguishes the Λ -electron particle species from the Λ' -electron particle species, by attributing a different essence to each. This might seem like too fine-grained a result to count as satisfying the Individuation Desiderata – and therefore, like an unattractive defense of realism against Ruetsche’s argument.

There are other costs of the Object Account. Like the Property Account, it might seem unparsimonious, since it posits infinitely many different particles. One might want a less profligate ontology. And like the Property Account, the Object Account does not provide a theory of exactly what unites all these different particles. In virtue of what do they resemble each other? In virtue of what are they roughly, but still aptly, called by the name ‘electron’?

The Object Account also violates the intuition that there is just one sort of electron. For according to the Object Account, there is a continuous infinity of numerically distinct Λ -electron particles. And even in the context of quantum physics, where bizarre and unintuitive physical posits are commonplace, a continuous infinity of distinct Λ -electron particles might be too strange to accept.

3.3 The Relation Account

In this section, I present the Relation Account of running couplings. According to the Relation Account, charge and mass are binary relations in which electrons stand to energy levels. So unlike in the Property Account, charge and mass are relational. And unlike in the Object Account, there is just one particle—the electron—which stands in these relations.

Here are the details. There is a binary determinable charge relation C with binary determinate charge relations c^r , and there is a binary determinable

mass relation M with binary determinate mass relations m^s .²⁰ As before, the superscripts ‘ r ’ and ‘ s ’ indicate that there is a charge determinate c^r of C , and a mass determinate m^s of M , for every single scalar quantity of charge r – and scalar quantity of mass s , respectively – which an object may have. Also, there are no energy level subscripts on C , M , the c^r , or the m^s ; and electrons like e have no energy level subscripts either. The reason: C , M , c^r , and m^s are two-place relations in which e stands to energy levels. C , M , c^r , and m^s are not one-place properties which e exemplifies; nor is there a separate particle for each energy level. Rather, C is the relation of being charged at, M is the relation of being massive at, each c^r is the relation of having r amount of charge at, each m^s is the relation of having s amount of mass at, and there is just one particle e which stands in these relations to energy levels below the cutoff. Charge and mass are not intrinsic properties of particles, but extrinsic relations in which particles stand.

Here is how the Relation Account can be used to describe facts about running charge and running mass. For each Λ below Λ_c , e stands in determinable C to Λ and e stands in determinable M to Λ . Formally: for each Λ below Λ_c , $C(e, \Lambda)$ and $M(e, \Lambda)$. In addition, for each Λ below Λ_c , e stands in the determinate c^r to Λ , where r is the quantity given by equation (Ch) corresponding to Λ , and e stands in the determinate m^s to Λ , where s is the quantity given by equation (Ma) corresponding to Λ . Formally, $c^r(e, \Lambda)$ and $m^s(e, \Lambda)$; note that $c^r(e, \Lambda)$ is the Relation Account’s regimentation of the ambiguous English phrase “The electron has r amount of charge at Λ ” discussed earlier. For any distinct Λ and Λ' below Λ_c , the determinate c^r charge relation – in which e stands to Λ – differs from the determinate $c^{r'}$ charge relation – in which e stands to Λ' – where r and r' are the quantities given by equation (Ch) that correspond to Λ and Λ' , respectively; and likewise for mass determinates. But there is nothing problematic about this. For c^r and $c^{r'}$, though different determinates of the same determinable C , obtain between different relata: c^r obtains between e and Λ , while $c^{r'}$ obtains between e and Λ' ; and again, likewise for mass. So the Relation Account avoids contradicting the relational generalization of the principle Preclusion.

The Relation Account has many of the benefits which the Property Account

²⁰For discussion of determinables and determinates in physics, see (French, 2014, p. 282).

has. For instance, the Relation Account addresses the issue raised by Fraser: the relations C , M , c^r , and m^s , are robustly realist and unfriendly to anti-realism, since they take—as relata—energies far above the energy level which experiments can probe. And all these posits are invariant across independent choices about how to model the physics at high energies.

In addition, the Relation Account satisfies the Individuation Desiderata, and so can be used to defend realism against Ruetsche’s argument. The defense endorses the following: determinables C and M are essential attributes of electrons, as are—for each Λ below Λ_c —determinates c_Λ^r and m_Λ^s for the specific r and s given by (Ch) and (Ma) corresponding to Λ . Given this view, it follows that for reasons entirely analogous to those discussed in Section 3.1, electrons are individuated—from neutrinos, quarks, and so on—by charge and mass.

Furthermore, the Relation Account does not require abandoning the standard realist tenet that scientific theories are true or false simpliciter. For the Relation Account captures the scale-relativity of facts about running couplings by analyzing charge and mass as relations which take energy levels as relata. So there is no need to index truth values to energy levels. The indexing shows up in the relata of the charge and mass relations.

Another potential benefit of the Relation Account: it is quite friendly to certain versions of ontic structural realism (McKenzie, 2017). The charge and mass relations posited by the Relation Account might be understood as among the structures which pluralist approaches to structural realism endorse (Glick, 2016, 2020). Whether or not this counts as a benefit, of course, depends on whether or not some version of ontic structural realism holds. But those in favor of ontic structural realism might like the relational, structuralist character of the Relation Account.

In addition, the Relation Account avoids the costs of both the Property Account and the Object Account. For starters, the Relation Account is more parsimonious: it posits a single binary determinable charge relation and a single binary determinable mass relation – and correspondingly a single collection of infinitely many different determinate charge relations and mass relations. That is far more parsimonious than the infinitely many charge and mass determinables, and infinitely many different collections of infinitely many corresponding deter-

minates, of the Property Account. It is also far more parsimonious than the infinitely many particles posited by the Object Account. In addition, the Relation Account—unlike the Property Account or the Object Account—supports a theory of what unites its posits. There is a single, unified, determinable charge relation; and there is a single, unified, determinable mass relation. Different instances of the determinable charge relation, obtaining between the electron and different energy levels, resemble each other in virtue of the fact that the very same relation C figures in those instances – and likewise for mass.

Nevertheless, the Relation Account has some costs of its own. For starters, it violates the intuition that charge and mass are one-place properties. Now, this might not be much of a cost. In my view, positing binary charge and mass relations is significantly less bizarre than, for instance, positing a continuous infinity of Λ -electron particles. But others might feel differently.

In addition, the Relation Account struggles to support Benitez’s bundle theory of effective particles. Bundle theories of objects often struggle to accommodate relations. It is somewhat difficult to understand what it would mean for a relation to be part of the bundle from which an object is composed.²¹

4 Conclusion

The following table summarizes the different posits, and different forms which facts about running couplings take, of the Property Account, the Object Account, and the Relation Account; note that Λ ranges over energies below Λ_c .

²¹Though see (Curtis, 2012) for an account of how relations might figure in bundles.

	Objects	Determinables	Determinates	Facts
Property Account	e	$C_\Lambda(\cdot), M_\Lambda(\cdot)$	$c_\Lambda^r(\cdot), m_\Lambda^s(\cdot)$	$C_\Lambda(e), M_\Lambda(e)$ $c_\Lambda^r(e), m_\Lambda^s(e)$
Object Account	e_Λ	$C(\cdot), M(\cdot)$	$c^r(\cdot), m^s(\cdot)$	$C(e_\Lambda), M(e_\Lambda)$ $c^r(e_\Lambda), m^s(e_\Lambda)$
Relation Account	e	$C(\cdot, \cdot), M(\cdot, \cdot)$	$c^r(\cdot, \cdot), m^s(\cdot, \cdot)$	$C(e, \Lambda), M(e, \Lambda)$ $c^r(e, \Lambda), m^s(e, \Lambda)$

The Relation Account strikes me as best, since it (i) addresses Fraser’s issue, (ii) combines nicely with realism about effective theories, (iii) satisfies the Individuation Desiderata and so supports realism against Ruetsche’s argument, (iv) might facilitate a bundle theory of effective particles, (v) upholds a standard semantic tenet of realism, (vi) is friendly to versions of ontic structural realism, (vii) is parsimonious, and (viii) provides a theory of what unites its ontological posits. But regardless, the three accounts jointly illustrate an important class of realist answers to difficult questions about running couplings in quantum field theories.

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